

ARROWSTACK CYBER SECURITY INTERNSHIP

Project 1 - Security Baseline Assessment Lab

Final Security Baseline Assessment Report

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Internship Domain	Cyber Security
Project	Security Baseline Assessment Lab
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Assessment Environment	Authorized personal Windows endpoint / lab environment

Prepared from the Arrowstack internship requirements and submitted assessment evidence.

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1. Executive Summary

This report documents a Security Baseline Assessment Lab completed as Project 1 of the Arrowstack one-month Cyber Security internship. The internship handbook defines Project 1 as a foundation/core industry task focused on assessing a deliberately simulated organization's endpoints, accounts and configurations against a security baseline, followed by evidence capture, risk prioritization, remediation planning and validation.

The submitted evidence shows a Windows endpoint assessed using PowerShell, Windows Event Viewer and Windows Update. The assessment covered operating-system identification, IPv4/network interfaces, local accounts, local Administrators membership, Windows Firewall profile status, Microsoft Defender status, Remote Desktop configuration, Security Event Log activity and Windows Update status.

Overall, the evidence demonstrates several positive baseline controls: Microsoft Defender and real-time protection were enabled, Windows Firewall profiles were enabled, Remote Desktop connections were disabled, and the built-in Administrator and Guest accounts were disabled. The assessment also identified items requiring review or remediation: pending Windows updates, an enabled local account named 'defaultuser100000', privileged membership of the primary Microsoft account, and firewall default actions reported as 'NotConfigured' in the captured command output. These observations should be treated as configuration-review findings rather than proof of compromise.

A Security Event Log screenshot also shows Event ID 5379 ('Credential Manager credentials were read'). The screenshot alone does not establish malicious activity; it is retained as evidence that security auditing is active and that credential-access-related events are being logged.

2. Project Context and Requirements

The Arrowstack handbook requires Project 1 to move from problem definition to a defensible result. The stated required activities are defined scope and authorization, asset/risk inventory, evidence capture, severity/risk prioritization and remediation validation.

- Define the target stakeholder, scope boundaries and success metrics before execution.
- Use a repeatable workflow with traceable decisions and organized artifacts.
- Handle realistic edge cases, invalid inputs, exceptions or uncertainty relevant to the domain.
- Include a validation/review step and retain evidence of the result.
- Document limitations and recommend realistic next improvements.

The required Project 1 deliverables are a scope statement, assessment evidence, risk register, remediation plan and final security report. The submitted report is structured around those requirements.

3. Scope, Authorization and Assessment Boundaries

Item	Assessment Definition
Target	One Windows endpoint used as the authorized assessment/lab system.
Assessment type	Security baseline review; configuration and evidence assessment.

Data sources	PowerShell command output, Windows Event Viewer and Windows Update screenshots.
Network scope	Locally visible network interfaces and configuration information on the endpoint.
Testing boundary	No external or unauthorized systems were tested.
Evidence principle	Findings are based only on captured evidence supplied for this report.
Success condition	Produce a reproducible baseline record, identify risks, propose remediation and retain evidence.

The Arrowstack handbook explicitly requires authorized/public/simulated systems and prohibits testing real systems without explicit written authorization. This assessment is therefore treated as an authorized personal endpoint/lab exercise.

4. Assessment Methodology

The assessment followed a practical baseline workflow:

1. Identify the endpoint operating system and host information.
2. Inventory IPv4 addresses, active interfaces and network configuration.
3. Review local user accounts and privileged Administrators group membership.
4. Review Windows Firewall profile status and captured default-action configuration.
5. Review Microsoft Defender antivirus, antispysware, real-time protection and service status.
6. Verify whether Remote Desktop connections are enabled or disabled.
7. Review the Windows Security Event Log for evidence of security auditing.
8. Review Windows Update status for pending updates.
9. Classify observations by baseline status and risk priority.
10. Develop remediation actions and validation steps.

5. Endpoint and Network Baseline

The captured PowerShell evidence identifies the following baseline characteristics.

Area	Observed Evidence	Baseline Interpretation
Operating system	Windows 11 Pro; WindowsVersion 2009; OS Build 26200; host DESKTOP- TQDKLGO	Documented endpoint version/build; update status requires follow-up.
Primary Wi-Fi	192.168.1.3; gateway/DNS 192.168.1.1; DHCP	Active Wi-Fi configuration captured.
Virtual networking	VMware VMnet8 192.168.74.1; VMware VMnet1 192.168.239.1	Virtual lab/network adapters present.
Tailscale	IPv4 100.109.228.83; Tailscale tunnel shown	Additional overlay/tunnel interface present and should be documented if required by

		the lab.
Other adapters	Several APIPA 169.254.x.x addresses and disconnected adapters	Likely disconnected/unconfigured interfaces; review unnecessary adapters as part of hardening.

The network evidence includes both active and inactive interfaces. The Wi-Fi interface is shown with a valid private IPv4 address and gateway, while several other interfaces are disconnected or have link-local/APIPA addresses. This is not by itself a vulnerability; it is an asset/configuration inventory observation.

6. Security Baseline Findings

ID	Finding	Priority	Evidence / Observation	Recommended Action
F-01	Pending Windows updates	Medium	Windows Update reports updates are available and presents Windows 11 version 26H2 as available.	Apply available updates after checking compatibility, backups and change impact. Record the resulting update state.
F-02	Enabled local account requiring review	Medium	The local-user inventory shows 'defaultuser100000' as Enabled=True with a historical last-logon timestamp.	Confirm whether the account is required. If unused or unauthorized, disable/remove it according to organizational policy and validate.
F-03	Privileged Microsoft account membership	Medium	The Administrators group contains the local Administrator account and the 'alman' Microsoft account.	Review whether everyday use requires administrative privileges. Prefer a standard account for routine work and separate elevation when practical.
F-04	Firewall default	Medium	Domain, Private and Public	Review effective

	actions not explicitly configured in captured output		profiles are Enabled=True, but DefaultInboundAction and DefaultOutboundAction are reported as NotConfigured.	firewall policy/rules. Confirm the intended inbound/outbound defaults and document the effective policy.
F-05	Remote Desktop disabled	Low / Positive Control	fDenyTSConnections is 1, indicating Remote Desktop connections are denied by this setting.	Retain the setting unless a documented business requirement exists; if enabled later, restrict access and document authorization.
F-06	Microsoft Defender Protection enabled	Low / Positive Control	AntivirusEnabled, AntispywareEnabled, RealTimeProtectionEnabled and AMServiceEnabled are all True.	Maintain the protection state and monitor definition/update health.
F-07	Security auditing evidence presents	Low / Positive Control	Security Event Log contains Windows security auditing entries, including Event ID 5379.	Retain appropriate auditing and periodically review high-value security events. Event 5379 alone is not proof of compromise.

7. Risk Register

The following risk register prioritizes configuration observations using qualitative likelihood/impact reasoning. These ratings are assessment judgments for this lab and are not claims that an incident occurred.

ID	Risk	Likelihood	Impact	Priority	Treatment
R-01	Unpatched endpoint may remain exposed to known weaknesses	Medium	Medium	Medium	Update and verify
R-02	Unused enabled	Medium	Medium	Medium	Review and

	account increases local attack surface				disable if unnecessary
R-03	Routine user account has administrator membership	Medium	High	Medium	Apply least privilege
R-04	Effective firewall defaults require confirmation	Medium	High	Medium	Validate effective policy
R-05	Unnecessary network interfaces/tunnels may increase exposure	Low	Medium	Low–Medium	Inventory and disable if unnecessary

8. Remediation Plan

Priority	Action	Owner	Validation Evidence	Status
1	Review and install available Windows updates; document final version/update state.	Endpoint owner	Windows Update screenshot + final OS/build check	Open
2	Review 'defaultuser100000'; disable/remove only if confirmed unnecessary.	Endpoint owner	Get-LocalUser output before/after	Open
3	Review whether the primary Microsoft account needs administrator rights.	Endpoint owner	Administrators group output after approved change	Open
4	Confirm effective Firewall inbound/outbound defaults and rules for all profiles.	Endpoint owner	Get-NetFirewallProfile + effective rule review	Open
5	Review disconnected/APIPA/virtual interfaces and retain only those required for the lab.	Endpoint owner	Updated interface inventory	Open
6	Continue security-event monitoring and investigate events that are unexpected in context.	Endpoint owner	Filtered Security Event Log evidence	Ongoing

9. Validation and Evidence Review

Validation was performed by checking that each major baseline area had captured evidence:

Control Area	Evidence Available	Validation Result
OS / host identification	PowerShell Get-ComputerInfo output	Validated
Network configuration	Get-NetIPAddress / Get-NetIPConfiguration outputs	Validated
Local users	Get-LocalUser output	Validated
Administrator membership	Get-LocalGroupMember Administrators output	Validated
Firewall	Get-NetFirewallProfile output	Validated; effective policy still needs confirmation
Endpoint protection	Get-MpComputerStatus output	Validated
Remote Desktop	Terminal Server registry value fDenyTSConnections=1	Validated as disabled by captured setting
Security auditing	Windows Security Event Log screenshot	Validated as present
Updates	Windows Update screenshot	

The validation process satisfies the internship requirement to retain evidence and document known limitations. The final state of remediation is not yet demonstrated by the supplied screenshots; therefore, remediation items are recorded as open unless the user performs and captures the recommended changes.

10. Limitations and Assumptions

- The assessment is based on the screenshots supplied for this report; no direct live inspection of the endpoint was performed.
- The evidence does not include a complete firewall-rule inventory, so 'NotConfigured' is not interpreted as proof that the firewall permits all traffic.
- Event ID 5379 indicates a Credential Manager credential-read event in the captured log; the screenshot alone does not establish malicious intent or compromise.
- The report does not claim that any identified configuration caused an incident.
- The presence of virtual adapters, Tailscale and APIPA addresses is treated as an inventory observation unless business/lab requirements show otherwise.
- The report does not independently verify whether every available Windows Update is security-related; it records the update state shown in the screenshot.

11. Conclusion

The Security Baseline Assessment established a documented security posture for the assessed Windows endpoint and produced the core evidence required by the Arrowstack Project 1 brief. The strongest observed controls were enabled endpoint protection, enabled Windows Firewall profiles, disabled Remote Desktop access and disabled built-in Administrator/Guest accounts. The main

follow-up areas are patch/update completion, local-account review, least-privilege review and confirmation of effective firewall policy.

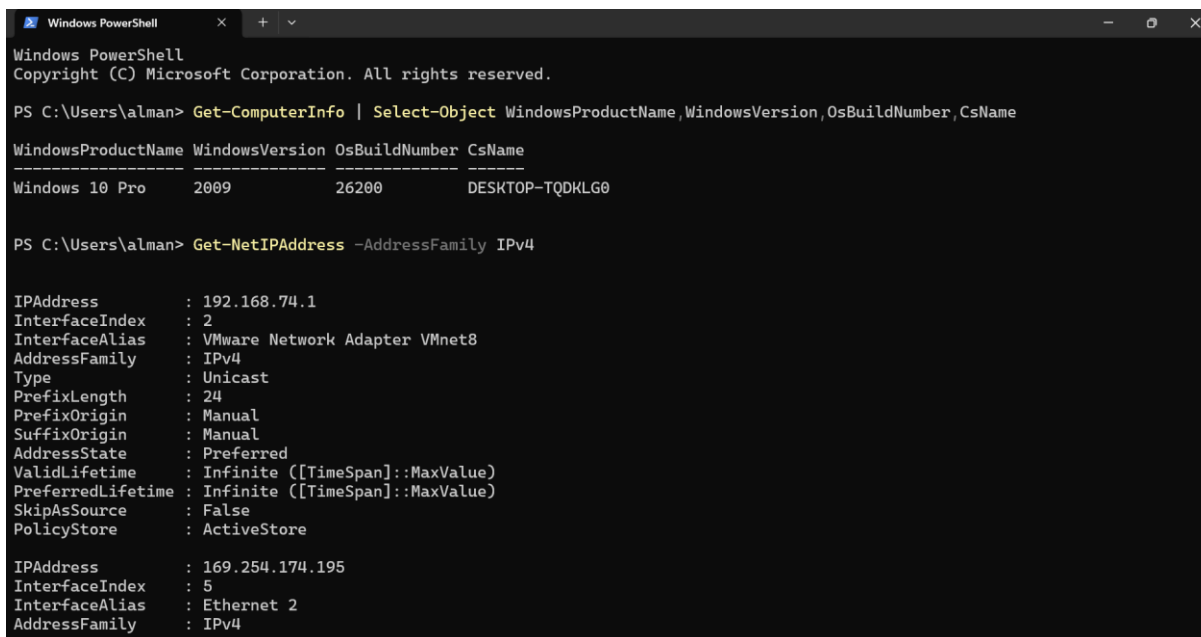
The assessment is suitable as a baseline because it records both positive controls and improvement areas, links observations to evidence and provides a remediation/validation path. The next step is to execute the open remediation items, capture before/after evidence and update the risk register to reflect the final validated state.

12. Deliverables Checklist

Deliverable	Status
Scope statement	Completed
Assessment evidence	Completed
Risk register	Completed
Remediation plan	Completed
Final security report	Completed

Appendix A — Evidence Screenshots

The following screenshots were supplied as assessment evidence. Captions describe the evidence without treating the screenshot alone as proof of malicious activity.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\alman> Get-ComputerInfo | Select-Object WindowsProductName,WindowsVersion,OsBuildNumber,CsName

-----
WindowsProductName  WindowsVersion  OsBuildNumber  CsName
-----
Windows 10 Pro      2009            26200          DESKTOP-TQDKLG0

PS C:\Users\alman> Get-NetIPAddress -AddressFamily IPv4

IPAddress           : 192.168.74.1
InterfaceIndex       : 2
InterfaceAlias       : VMware Network Adapter VMnet8
AddressFamily        : IPv4
Type                 : Unicast
PrefixLength         : 24
PrefixOrigin         : Manual
SuffixOrigin         : Manual
AddressState         : Preferred
ValidLifetime        : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime    : Infinite ([TimeSpan]::MaxValue)
SkipAsSource         : False
PolicyStore          : ActiveStore

IPAddress           : 169.254.174.195
InterfaceIndex       : 5
InterfaceAlias       : Ethernet 2
AddressFamily        : IPv4
```

Evidence 1 — Operating system and initial IPv4 address inventory

```
Windows PowerShell
IPAddress      : 169.254.151.32
InterfaceIndex : 26
InterfaceAlias : Ethernet
AddressFamily  : IPv4
Type           : Unicast
PrefixLength   : 16
PrefixOrigin   : WellKnown
SuffixOrigin    : Link
AddressState    : Tentative
ValidLifetime   : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource    : False
PolicyStore     : ActiveStore

IPAddress      : 169.254.208.46
InterfaceIndex : 24
InterfaceAlias  : Local Area Connection* 9
AddressFamily   : IPv4
Type            : Unicast
PrefixLength     : 16
PrefixOrigin     : WellKnown
SuffixOrigin      : Link
AddressState      : Tentative
ValidLifetime     : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource      : False
PolicyStore       : ActiveStore

IPAddress      : 169.254.143.237
InterfaceIndex  : 6
InterfaceAlias   : Bluetooth Network Connection
AddressFamily    : IPv4
Type             : Unicast
```

Evidence 2 — Additional IPv4 interface inventory

```
Windows PowerShell
PolicyStore      : ActiveStore

IPAddress        : 169.254.80.110
InterfaceIndex   : 4
InterfaceAlias   : Cellular
AddressFamily    : IPv4
Type             : Unicast
PrefixLength     : 16
PrefixOrigin     : WellKnown
SuffixOrigin     : Link
AddressState     : Tentative
ValidLifetime    : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource     : False
PolicyStore      : ActiveStore

IPAddress        : 192.168.1.3
InterfaceIndex   : 20
InterfaceAlias   : Wi-Fi
AddressFamily    : IPv4
Type             : Unicast
PrefixLength     : 24
PrefixOrigin     : Dhcp
SuffixOrigin     : Dhcp
AddressState     : Preferred
ValidLifetime    : 20:53:59
PreferredLifetime : 20:53:59
SkipAsSource     : False
PolicyStore      : ActiveStore

IPAddress        : 100.109.228.83
```

Evidence 3 — Wi-Fi, cellular and Tailscale IPv4 information

```
Windows PowerShell
SkipAsSource      : False
PolicyStore       : ActiveStore

IPAddress         : 169.254.174.195
InterfaceIndex    : 5
InterfaceAlias    : Ethernet 2
AddressFamily     : IPv4
Type              : Unicast
PrefixLength      : 16
PrefixOrigin      : WellKnown
SuffixOrigin      : Link
AddressState      : Tentative
ValidLifetime     : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource      : False
PolicyStore       : ActiveStore

IPAddress         : 192.168.239.1
InterfaceIndex    : 31
InterfaceAlias    : VMware Network Adapter VMnet1
AddressFamily     : IPv4
Type              : Unicast
PrefixLength      : 24
PrefixOrigin      : Manual
SuffixOrigin      : Manual
AddressState      : Preferred
ValidLifetime     : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource      : False
PolicyStore       : ActiveStore
```

Evidence 4 — VMware VMnet1 and additional interface information

```

PS C:\Users\alman> Get-NetIPConfiguration

InterfaceAlias      : VMware Network Adapter VMnet8
InterfaceIndex      : 2
InterfaceDescription : VMware Virtual Ethernet Adapter for VMnet8
IPv4Address          : 192.168.74.1
IPv6DefaultGateway   :
IPv4DefaultGateway   :
DNSServer            : fec0:0:0:ffff::1
                    : fec0:0:0:ffff::2
                    : fec0:0:0:ffff::3

InterfaceAlias      : VMware Network Adapter VMnet1
InterfaceIndex      : 31
InterfaceDescription : VMware Virtual Ethernet Adapter for VMnet1
IPv4Address          : 192.168.239.1
IPv6DefaultGateway   :
IPv4DefaultGateway   :
DNSServer            : fec0:0:0:ffff::1
                    : fec0:0:0:ffff::2
                    : fec0:0:0:ffff::3

InterfaceAlias      : Tailscale
InterfaceIndex      : 13
InterfaceDescription : Tailscale Tunnel
NetProfile.Name      : Tailscale
IPv6Address          : fd7a:115c:a1e0::2636:e454
IPv4Address          : 100.109.228.83
IPv6DefaultGateway   :
IPv4DefaultGateway   :

```

Evidence 5 — Network configuration and interface configuration

```
Windows PowerShell
DNSServer      : fec0:0:0:ffff::1
                fec0:0:0:ffff::2
                fec0:0:0:ffff::3

InterfaceAlias : Wi-Fi
InterfaceIndex : 20
InterfaceDescription : Intel(R) Dual Band Wireless-AC 8265
NetProfile.Name  : Hassan.Mujtaba
IPv4Address      : 192.168.1.3
IPv4DefaultGateway : 192.168.1.1
DNSServer       : 192.168.1.1

InterfaceAlias : Ethernet 2
InterfaceIndex : 5
InterfaceDescription : TAP-Surfshark Windows Adapter V9
NetAdapter.Status : Disconnected

InterfaceAlias : Cellular
InterfaceIndex : 4
InterfaceDescription : Fibocom L830-EB
NetAdapter.Status : Disconnected

InterfaceAlias : Bluetooth Network Connection
InterfaceIndex : 6
InterfaceDescription : Bluetooth Device (Personal Area Network)
NetAdapter.Status : Disconnected

InterfaceAlias : Ethernet
InterfaceIndex : 26
InterfaceDescription : Intel(R) Ethernet Connection (4) I219-V
NetAdapter.Status : Disconnected
```

Evidence 6 — Active Wi-Fi configuration and disconnected adapters

```
Windows PowerShell
InterfaceAlias      : Cellular
InterfaceIndex      : 4
InterfaceDescription : Fibocom L830-EB
NetAdapter.Status   : Disconnected

InterfaceAlias      : Bluetooth Network Connection
InterfaceIndex      : 6
InterfaceDescription : Bluetooth Device (Personal Area Network)
NetAdapter.Status   : Disconnected

InterfaceAlias      : Ethernet
InterfaceIndex      : 26
InterfaceDescription : Intel(R) Ethernet Connection (4) I219-V
NetAdapter.Status   : Disconnected

PS C:\Users\alman> Get-LocalUser | Select Name, Enabled, LastLogon

Name                Enabled LastLogon
----                -
Administrator       False
alman                True
DefaultAccount       False
defaultuser100000    True 08/08/2025 16:04:32
Guest                False
WDAGUtilityAccount   False
WsiAccount           False 25/02/2026 14:40:09

PS C:\Users\alman>
```

Evidence 7 — Local user account inventory

```
PS C:\Users\alman> Get-LocalGroupMember Administrators

ObjectClass Name                                PrincipalSource
-----
User          DESKTOP-TQDKLG0\Administrator Local
User          DESKTOP-TQDKLG0\alman      MicrosoftAccount

PS C:\Users\alman> |
```

Evidence 8 — Local Administrators group membership

```
Windows PowerShell
PS C:\Users\alman> Get-NetFirewallProfile | Select Name, Enabled, DefaultInboundAction, DefaultOutboundAction

Name      Enabled DefaultInboundAction DefaultOutboundAction
-----
Domain    True     NotConfigured         NotConfigured
Private   True     NotConfigured         NotConfigured
Public    True     NotConfigured         NotConfigured

PS C:\Users\alman> Get-MpComputerStatus | Select AntivirusEnabled, AntispywareEnabled, RealTimeProtectionEnabled, AMServiceEnabled

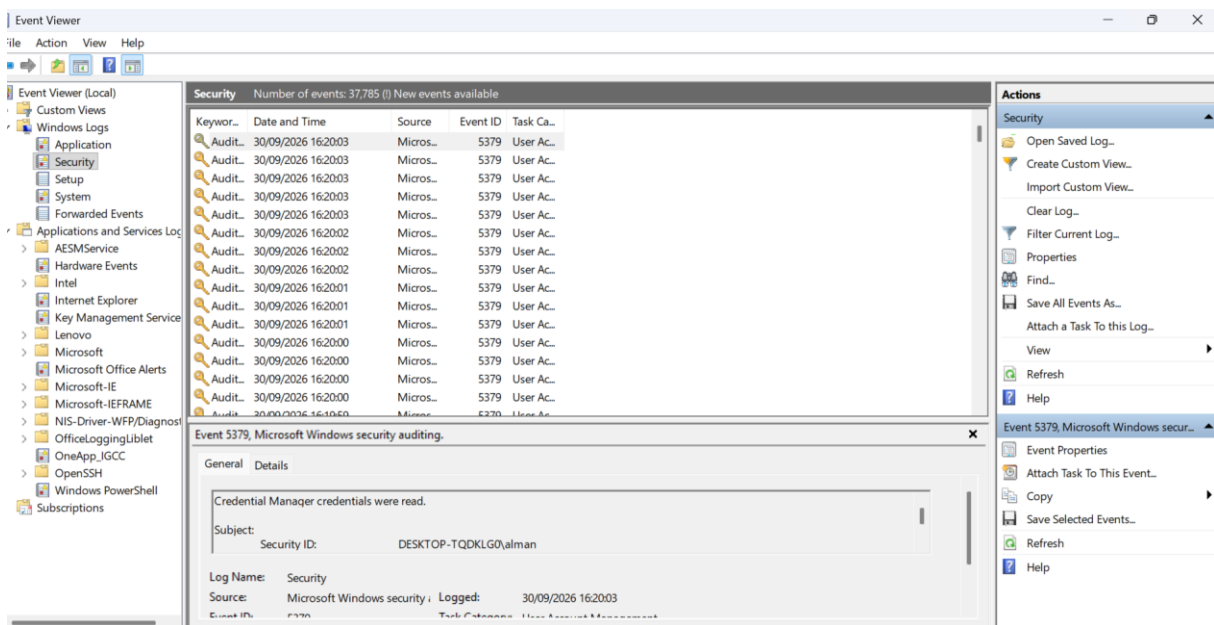
AntivirusEnabled AntispywareEnabled RealTimeProtectionEnabled AMServiceEnabled
-----
True             True                True                    True

PS C:\Users\alman> Get-ItemProperty "HKLM:\System\CurrentControlSet\Control\Terminal Server" -Name fDenyTSConnections

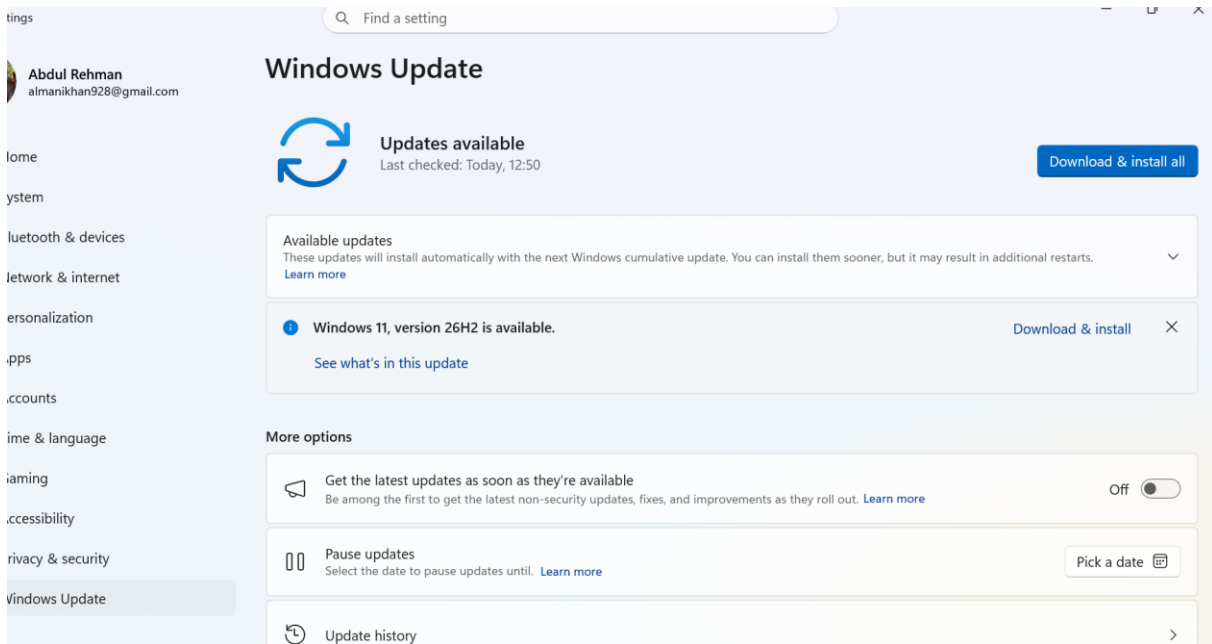
fDenyTSConnections : 1
PSPath              : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\Terminal
                    Server
PSParentPath         : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control
PSParentPath         : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control
PSChildName          : Terminal Server
PSDrive              : HKLM
PSProvider            : Microsoft.PowerShell.Core\Registry

PS C:\Users\alman>
```

Evidence 9 — Firewall, Microsoft Defender and RDP configuration



Evidence 10 — Windows Security Event Log evidence



Evidence 11 — Windows Update status